Soft constraint continuation: Cantera vs NN | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

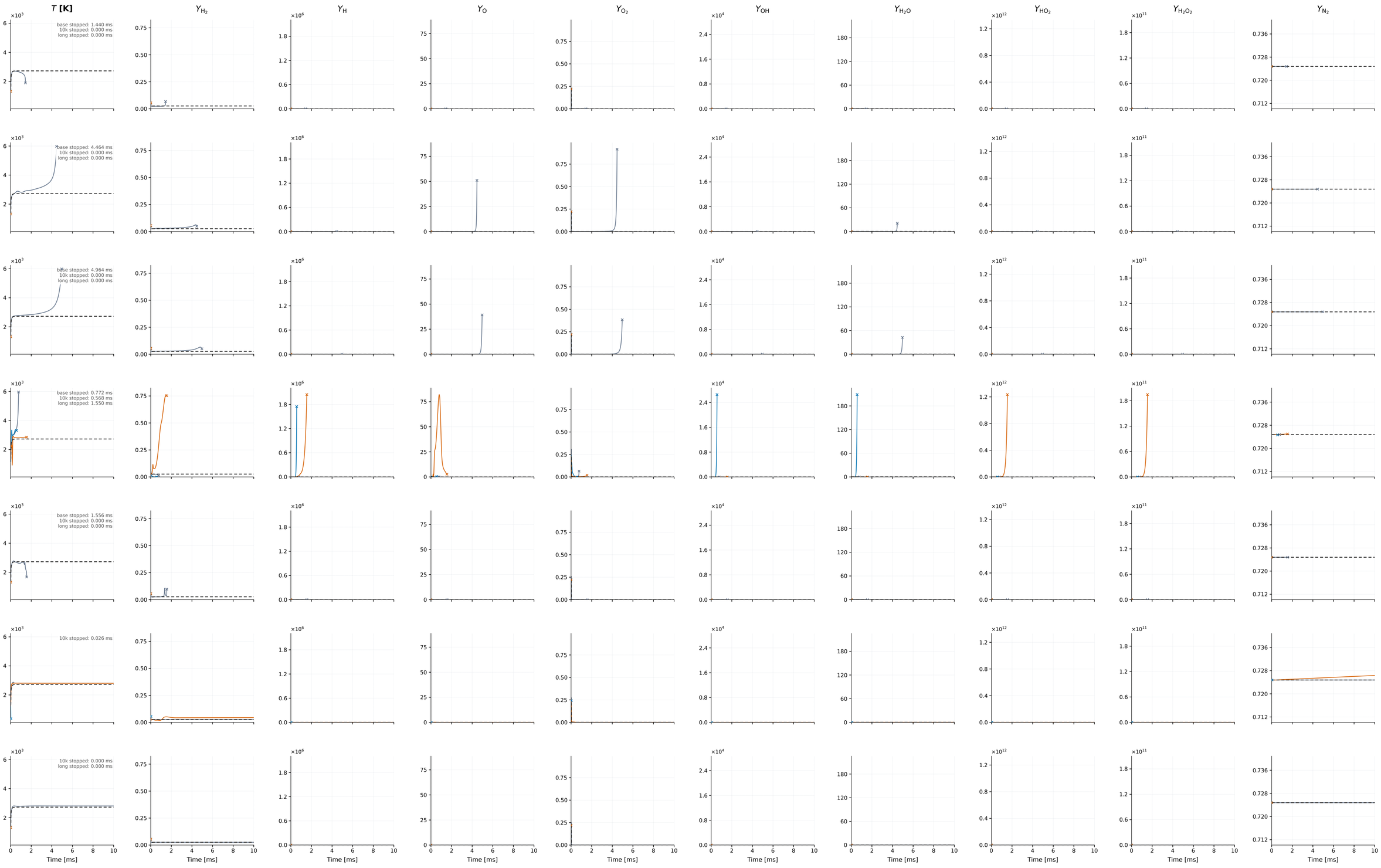
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 04 | NRMSE (util.py)

T0 = 1317.5 K P0 = 1.244 atm phi0 = 1.984 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.440 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 4.464 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 4.964 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.772 ms**
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.568 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 1.550 ms**
5 ANN10 + T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.556 ms**
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	baseline	0.0309	0.0188	0.215	0.559	0.164	0.406	0.0425	0.406	0.363	6.13e-16	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.026 ms**
6 T/P/ϕ DeepONet10	softlong	0.031	0.632	2.12e+03	48	0.131	1.49	0.569	461	158	1.23e-03	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.0296	0.0176	0.223	0.634	0.106	0.427	0.047	0.294	0.389	1.53e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 05

T_0 = 1049.0 K P_0 = 4.666 atm ϕ_0 = 0.674 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K]: base stopped: 1.660 ms; 10k stopped: 0.000 ms; long stopped: 0.020 ms

Y_{H_2}

Y_H

Y_O

Y_{O_2}

Y_{OH}

Y_{H_2O}

Y_{HO_2}

$Y_{H_2O_2}$

Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 05 | NRMSE (util.py)

T0 = 1049.0 K P0 = 4.666 atm phi0 = 0.674 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.660 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.020 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.186 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.002 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.544 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.004 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.420 ms**
4 T/P DeepONet10	soft10k	0.669	0.986	1.48e+03	1.03e+14	4.87e+07	4.04e+15	8.00e+09	5.05e+22	5.88e+28	5.10e-04	Complete
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.190 ms**
5 ANN10 + T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.950 ms**
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.162 ms**
6 T/P/ϕ DeepONet10	baseline	1.71	0.835	29.8	1.22e+03	16.9	336	0.817	2.37	0.961	1.48e-16	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 1.122 ms**
6 T/P/ϕ DeepONet10	softlong	0.541	0.981	4.33e+29	2.64e+28	2.40e+11	4.39e+25	1	1.57e+30	5.97e+27	1.26e-03	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	1.5	0.903	68.8	2.19e+03	16.8	203	0.602	2.85	0.951	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.390 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 07

T_0 = 930.6 K P_0 = 3.722 atm ϕ_0 = 0.754 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

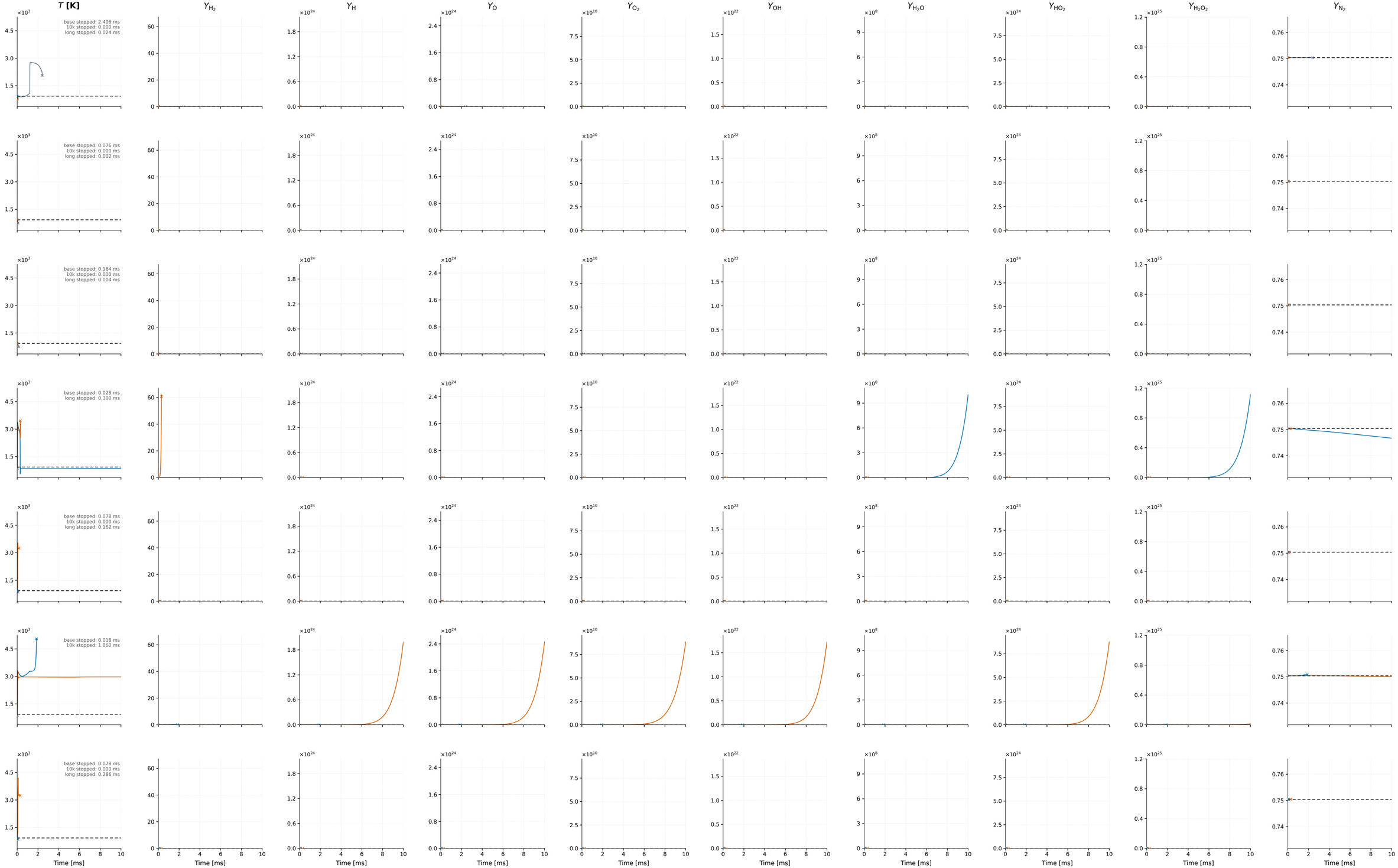
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 07 | NRMSE (util.py)

T0 = 930.6 K P0 = 3.722 atm phi0 = 0.754 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 2.406 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.024 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.076 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.002 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.164 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.004 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.028 ms**
4 T/P DeepONet10	soft10k	0.368	0.996	1.76e+09	1.62e+14	2.02e+05	1.57e+24	1.23e+15	4.06e+24	6.11e+30	2.67e-03	Complete
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.300 ms**
5 ANN10 + T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.078 ms**
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.162 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 1.860 ms**
6 T/P/ϕ DeepONet10	softlong	2.18	0.987	1.20e+35	7.12e+34	8.78e+10	8.22e+32	3.35e+04	3.90e+30	6.93e+28	1.65e-04	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.078 ms**
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.286 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 18

T_0 = 1069.1 K P_0 = 2.170 atm ϕ_0 = 1.544 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

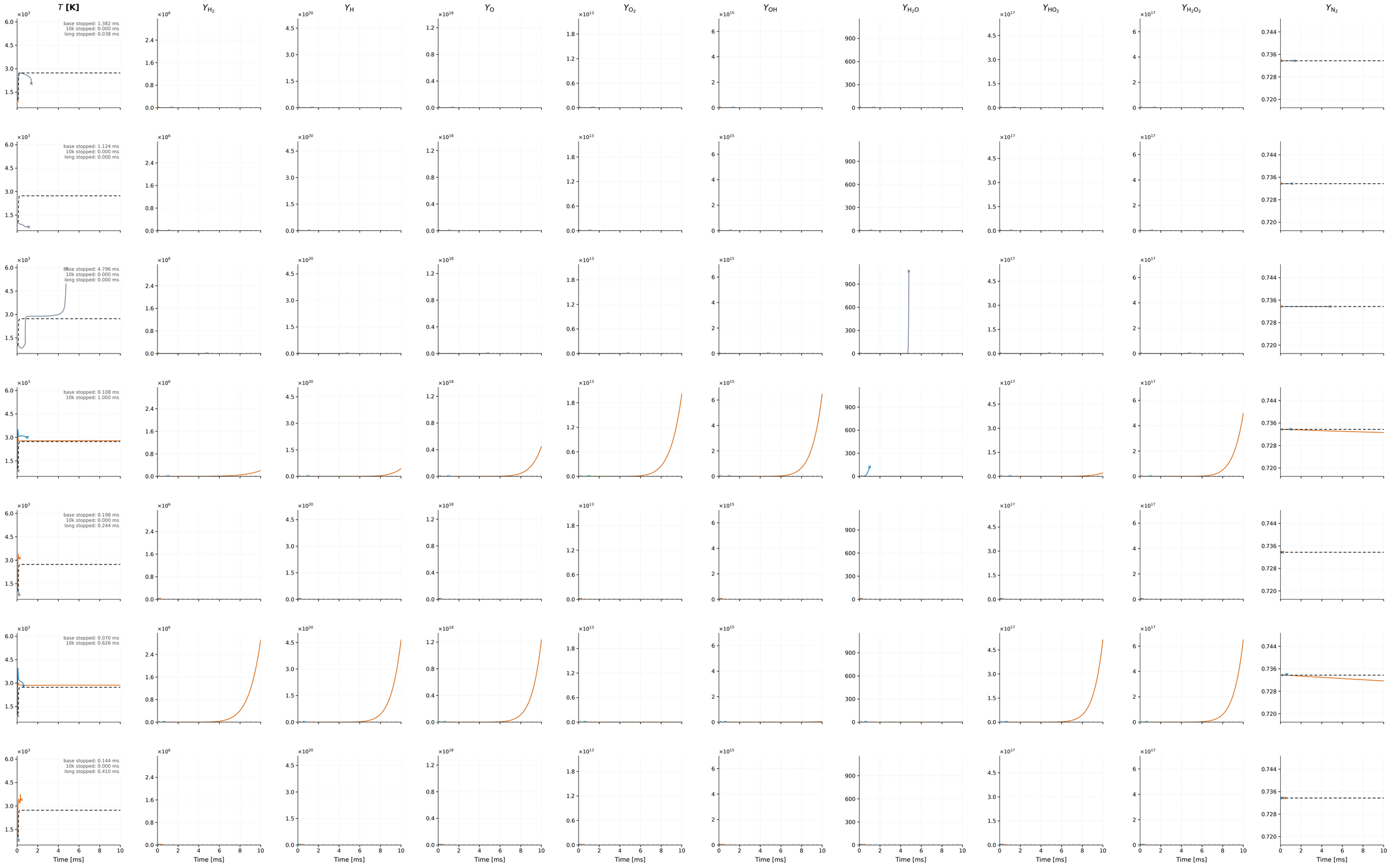
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 18 | NRMSE (util.py)

T0 = 1069.1 K P0 = 2.170 atm phi0 = 1.544 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

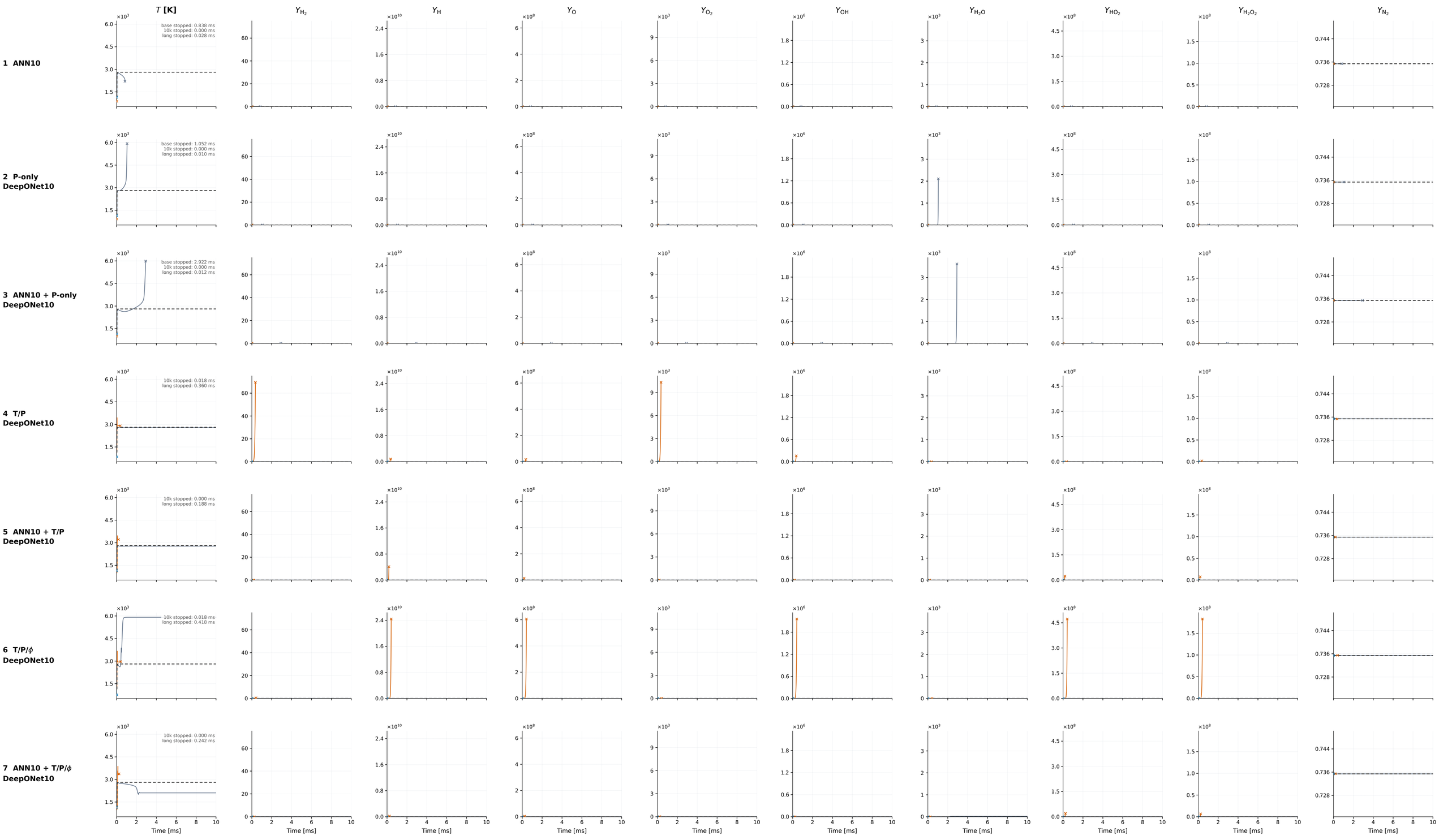
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.382 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.038 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.124 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 4.796 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.108 ms**
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 1.000 ms**
4 T/P DeepONet10	softlong	0.0662	3.59e+06	1.15e+22	1.97e+20	1.83e+14	2.12e+17	0.995	3.38e+20	9.39e+22	8.93e-04	Complete
5 ANN10 + T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.198 ms**
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.244 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.070 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.626 ms**
6 T/P/ϕ DeepONet10	softlong	0.0823	4.34e+07	1.18e+23	5.20e+20	1.69e+07	1.22e+15	1	7.57e+21	1.20e+23	1.59e-03	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.144 ms**
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.410 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 32

T_0 = 1140.7 K P_0 = 3.354 atm ϕ_0 = 1.463 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 32 | NRMSE (util.py)

T0 = 1140.7 K P0 = 3.354 atm phi0 = 1.463 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

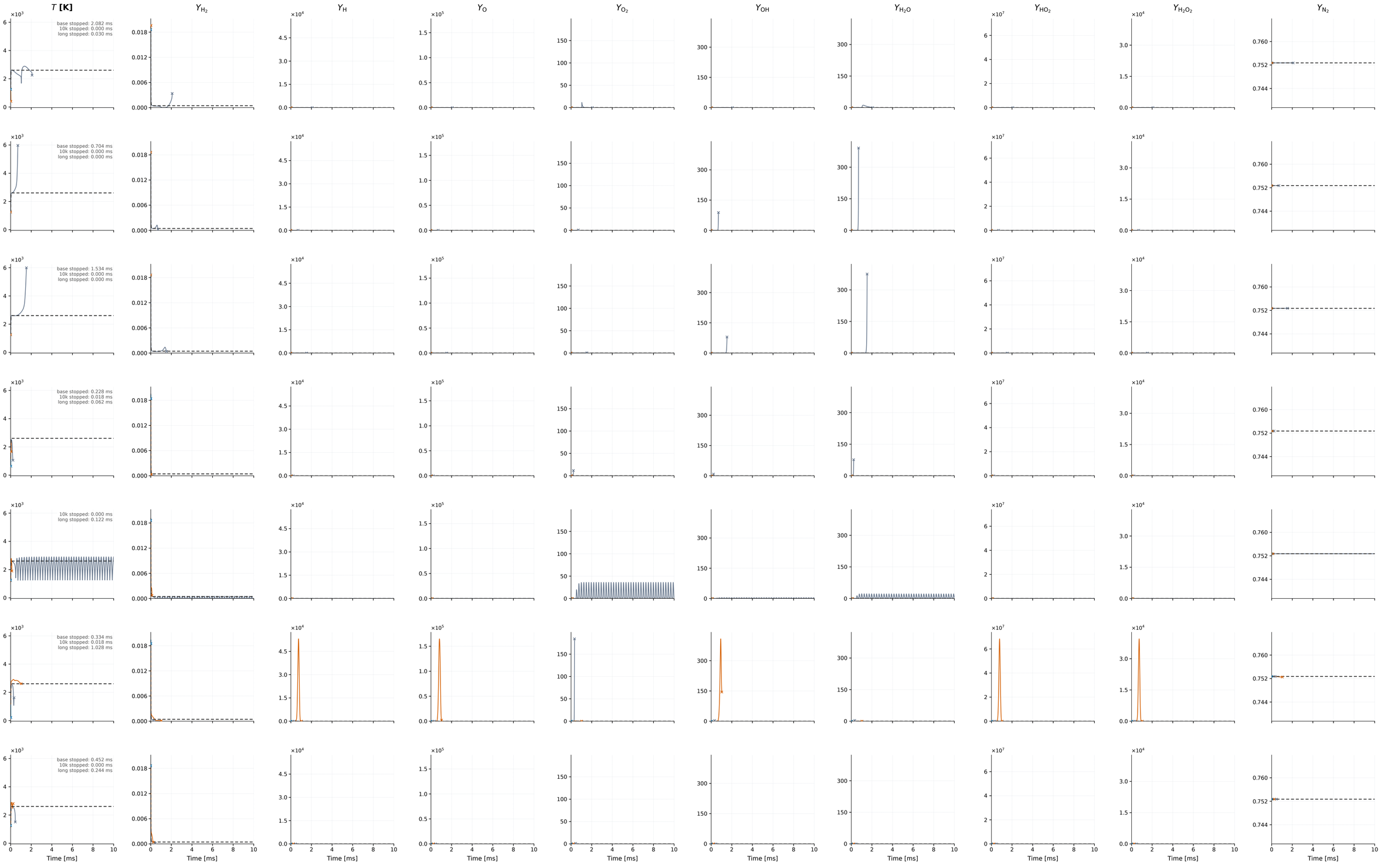
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.838 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.028 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.052 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.010 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 2.922 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.012 ms**
4 T/P DeepONet10	baseline	0.0202	0.0859	0.158	0.382	0.588	0.0956	0.0328	1.22	0.997	1.06e-15	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.360 ms**
5 ANN10 + T/P DeepONet10	baseline	0.0163	0.0579	0.204	0.378	0.395	0.172	0.0257	1.12	1.1	1.06e-15	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.188 ms**
6 T/P/ϕ DeepONet10	baseline	1.07	1.94	2.39	2.27e+03	58	3.86e+03	0.953	23.8	421	1.06e-15	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
6 T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.418 ms**
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.224	2.05	102	0.727	0.326	0.829	75.3	1.85	30.6	1.06e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.242 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 39

T_0 = 1250.1 K P_0 = 2.297 atm ϕ_0 = 0.646 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 39 | NRMSE (util.py)

T0 = 1250.1 K P0 = 2.297 atm phi0 = 0.646 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 2.082 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.030 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.704 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.534 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.228 ms**
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.062 ms**
5 ANN10 + T/P DeepONet10	baseline	0.225	0.427	126	12	214	167	67.7	97.8	29.2	5.90e-16	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.122 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.334 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
6 T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 1.028 ms**
7 ANN10 + T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.452 ms**
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.244 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 44

T_0 = 1031.1 K P_0 = 2.272 atm ϕ_0 = 0.770 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K]
base stopped: 2.236 ms
10k stopped: 0.000 ms
long stopped: 0.036 ms

Y_{H_2}

Y_H

Y_O

Y_{O_2}

Y_{OH}

Y_{H_2O}

Y_{HO_2}

$Y_{H_2O_2}$

Y_{N_2}

T [K]
base stopped: 3.912 ms
10k stopped: 0.000 ms
long stopped: 0.010 ms

Y_{H_2}

Y_H

Y_O

Y_{O_2}

Y_{OH}

Y_{H_2O}

Y_{HO_2}

$Y_{H_2O_2}$

Y_{N_2}

T [K]
base stopped: 3.514 ms
10k stopped: 0.000 ms
long stopped: 0.020 ms

Y_{H_2}

Y_H

Y_O

Y_{O_2}

Y_{OH}

Y_{H_2O}

Y_{HO_2}

$Y_{H_2O_2}$

Y_{N_2}

T [K]
base stopped: 0.414 ms
10k stopped: 1.314 ms
long stopped: 8.738 ms

Y_{H_2}

Y_H

Y_O

Y_{O_2}

Y_{OH}

Y_{H_2O}

Y_{HO_2}

$Y_{H_2O_2}$

Y_{N_2}

T [K]
base stopped: 0.870 ms
10k stopped: 0.000 ms
long stopped: 0.268 ms

Y_{H_2}

Y_H

Y_O

Y_{O_2}

Y_{OH}

Y_{H_2O}

Y_{HO_2}

$Y_{H_2O_2}$

Y_{N_2}

T [K]
base stopped: 0.374 ms
10k stopped: 0.650 ms
long stopped: 0.838 ms

Y_{H_2}

Y_H

Y_O

Y_{O_2}

Y_{OH}

Y_{H_2O}

Y_{HO_2}

$Y_{H_2O_2}$

Y_{N_2}

T [K]
10k stopped: 0.000 ms
long stopped: 0.350 ms

Y_{H_2}

Y_H

Y_O

Y_{O_2}

Y_{OH}

Y_{H_2O}

Y_{HO_2}

$Y_{H_2O_2}$

Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 44 | NRMSE (util.py)

T0 = 1031.1 K P0 = 2.272 atm phi0 = 0.770 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 2.236 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.036 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 3.912 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.010 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 3.514 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.020 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.414 ms**
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 1.314 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 8.738 ms**
5 ANN10 + T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.870 ms**
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.268 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.374 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.650 ms**
6 T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.838 ms**
7 ANN10 + T/P/ϕ DeepONet10	baseline	1.21	1.51	31.7	859	15.7	160	0.333	6.13	5.11	2.22e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.350 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 45

T_0 = 1181.9 K P_0 = 4.842 atm ϕ_0 = 1.099 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

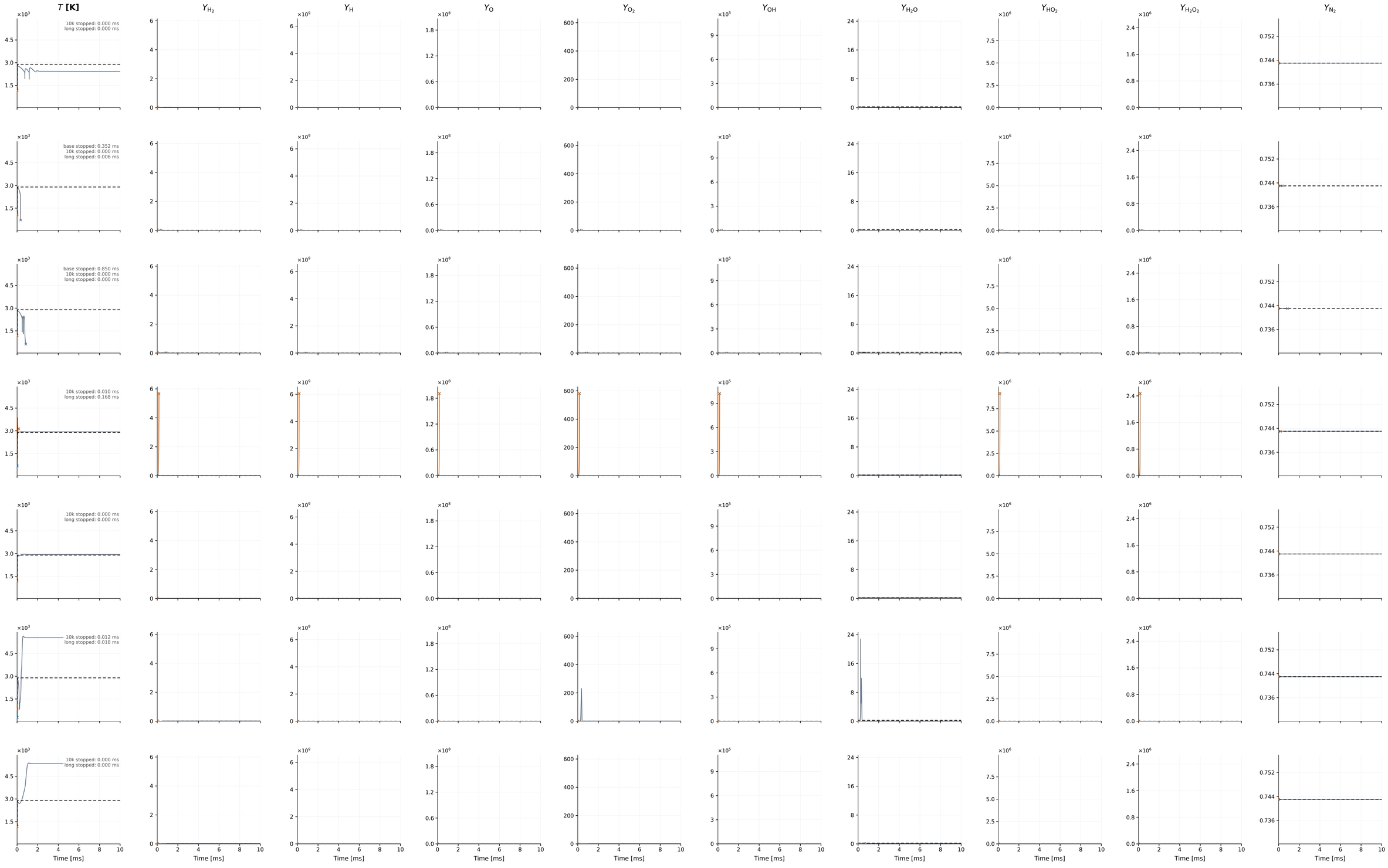
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 45 | NRMSE (util.py)

T0 = 1181.9 K P0 = 4.842 atm phi0 = 1.099 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.154	1.63	0.954	0.964	0.621	0.93	0.638	0.705	0.788	1.49e-16	Complete
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.352 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.006 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.850 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	baseline	0.0217	0.262	0.261	0.199	0.511	0.0576	0.0293	0.84	0.605	1.49e-16	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.010 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.168 ms**
5 ANN10 + T/P DeepONet10	baseline	0.0214	0.315	0.312	0.16	0.505	0.116	0.0315	0.885	0.653	1.49e-16	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	baseline	0.897	2.9	2.57	818	1.26e+03	788	5.08	31.5	155	1.49e-16	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.012 ms**
6 T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.805	3.6	3.74	1.03e+03	80.1	212	0.391	12.5	67.8	1.49e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 53

T_0 = 1257.8 K P_0 = 2.103 atm ϕ_0 = 1.799 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

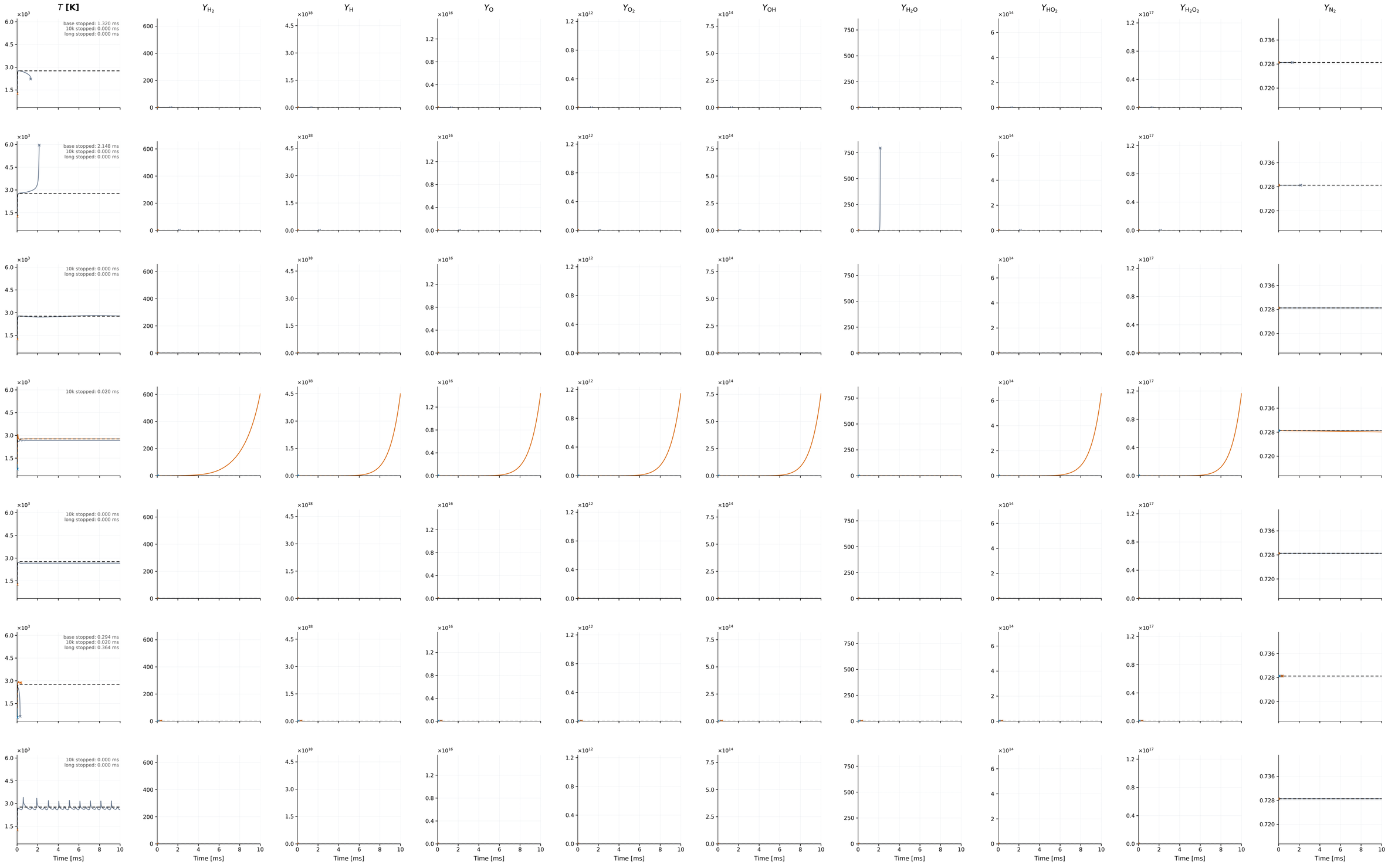
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 53 | NRMSE (util.py)

T0 = 1257.8 K P0 = 2.103 atm phi0 = 1.799 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

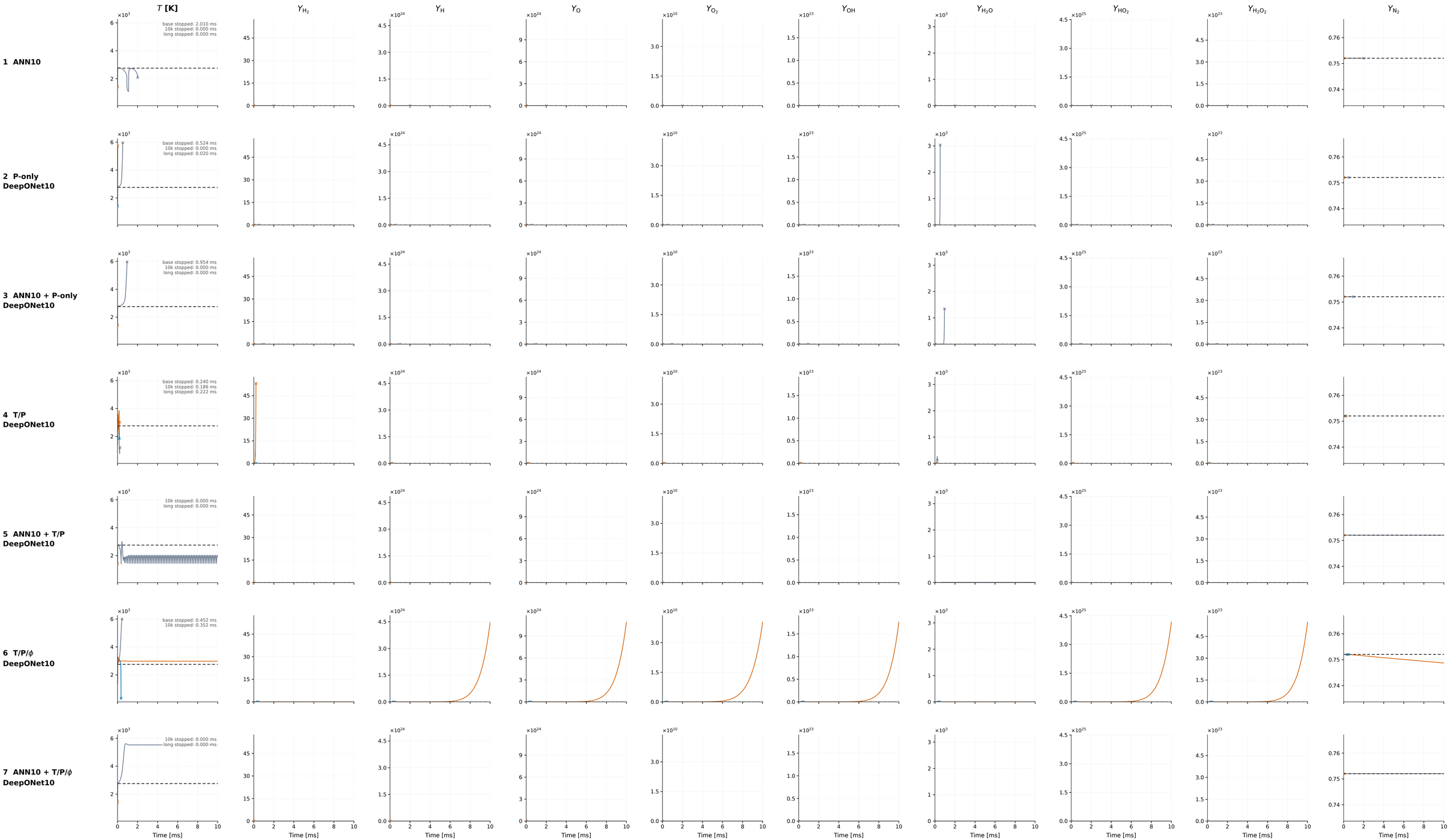
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.320 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 2.148 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	baseline	0.0124	0.292	0.396	0.464	0.0498	0.204	0.136	0.121	0.506	7.62e-16	Complete
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	baseline	0.0377	0.0241	0.286	0.477	0.312	0.332	0.0163	0.592	0.456	7.62e-16	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.020 ms**
4 T/P DeepONet10	softlong	0.0111	7.92e+03	8.29e+20	6.51e+18	2.73e+13	2.45e+16	0.985	2.90e+19	1.25e+23	3.34e-04	Complete
5 ANN10 + T/P DeepONet10	baseline	0.0356	0.0124	0.425	0.598	0.201	0.426	0.106	0.415	0.373	7.62e-16	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.294 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.020 ms**
6 T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.364 ms**
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.05	0.249	0.583	0.893	0.258	0.86	0.241	0.547	1.1	7.62e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 57

T_0 = 1412.0 K P_0 = 4.083 atm ϕ_0 = 0.677 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 57 | NRMSE (util.py)

T0 = 1412.0 K P0 = 4.083 atm phi0 = 0.677 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 2.010 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.524 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.020 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.954 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.240 ms**
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.186 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.222 ms**
5 ANN10 + T/P DeepONet10	baseline	0.332	0.813	56.4	0.932	63.3	136	59.5	57.4	32.6	1.48e-16	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.452 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.352 ms**
6 T/P/ϕ DeepONet10	softlong	0.0824	5.13	7.55e+27	7.17e+26	1.34e+11	2.57e+24	0.999	3.40e+29	7.23e+28	2.50e-03	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.977	9.31	18.2	665	32.5	95.2	0.146	6.37	26.8	1.48e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 64

T_0 = 1045.3 K P_0 = 4.301 atm ϕ_0 = 0.595 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

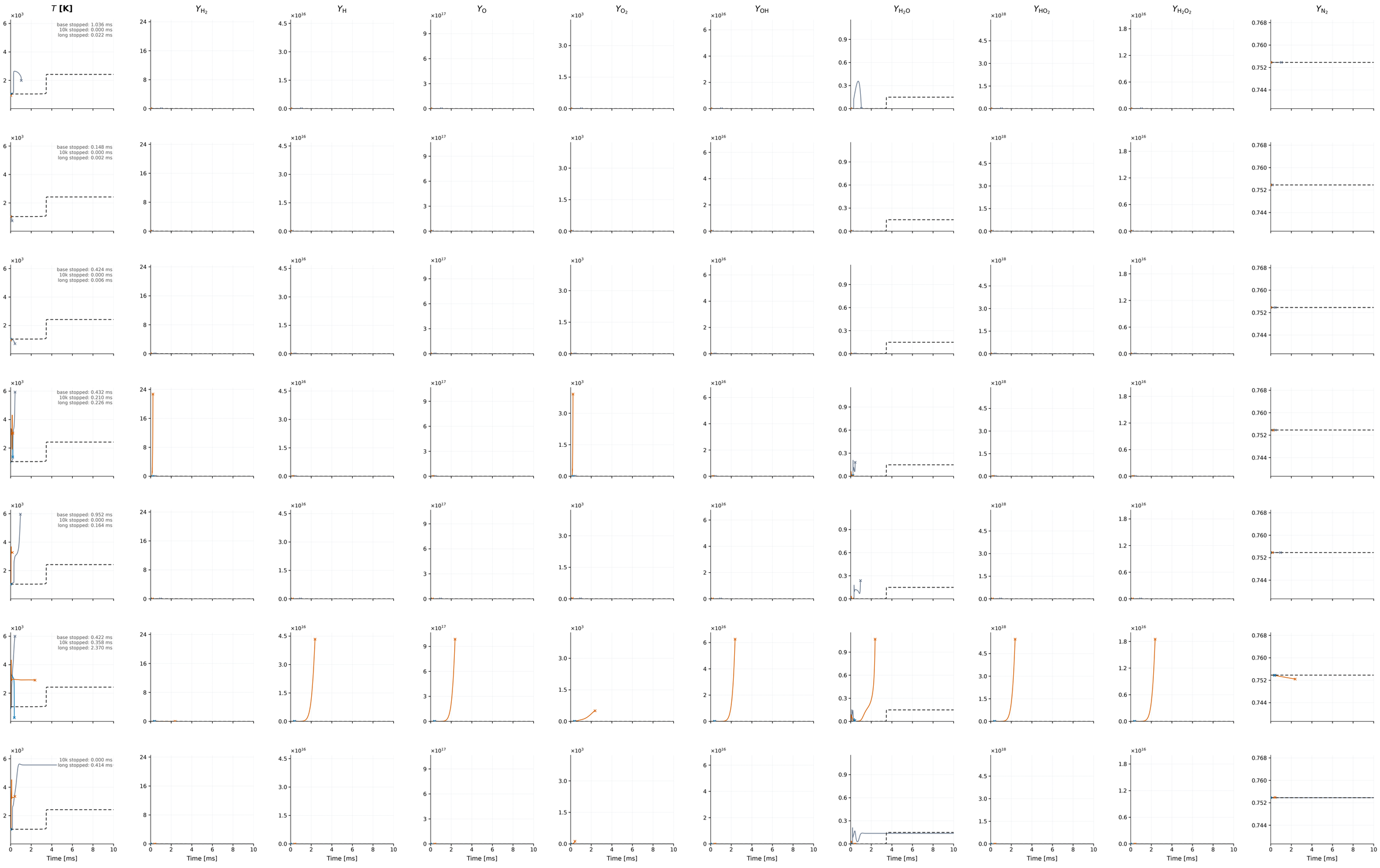
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 64 | NRMSE (util.py)

T0 = 1045.3 K P0 = 4.301 atm phi0 = 0.595 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

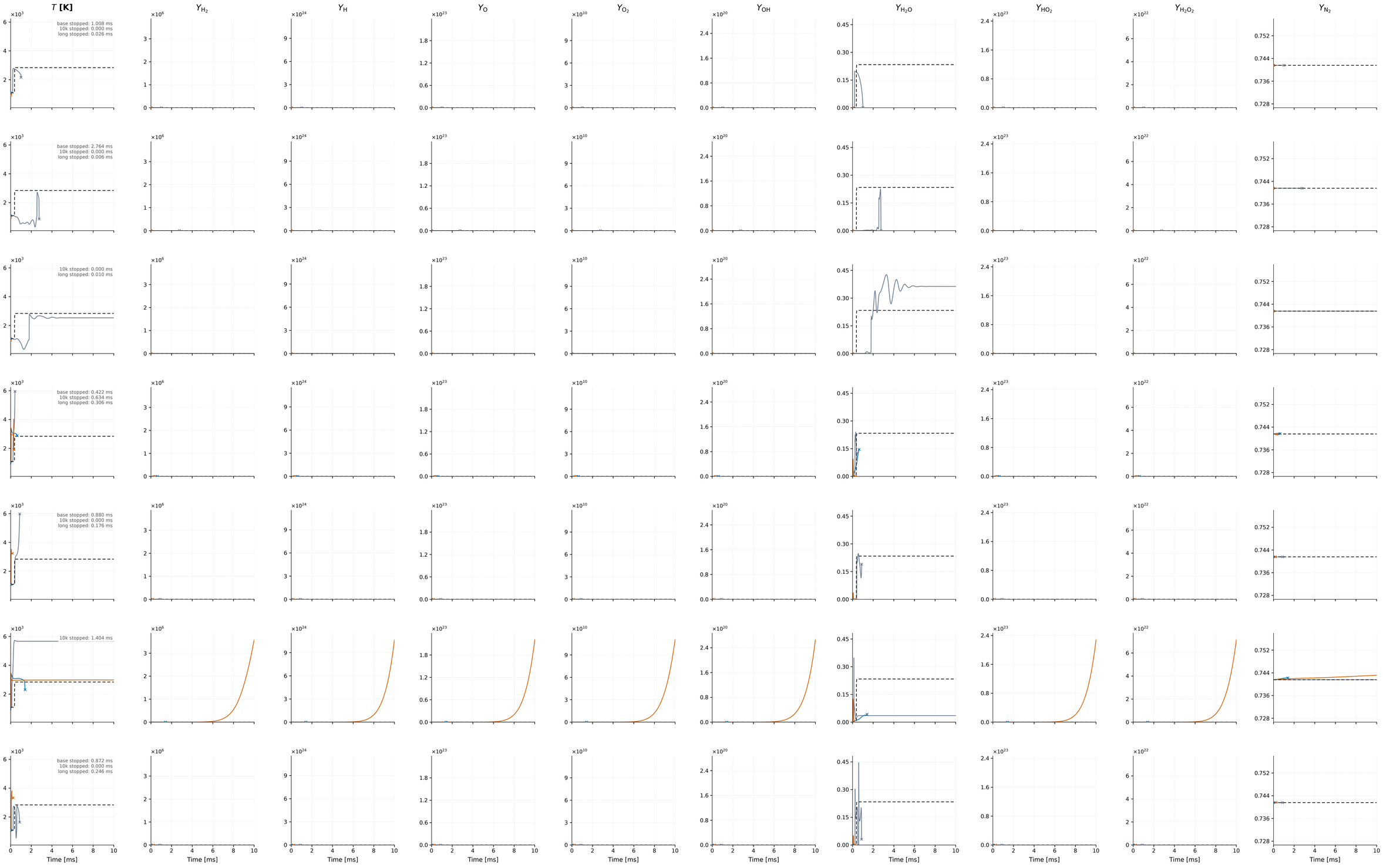
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.036 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.022 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.148 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.002 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.424 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.006 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.432 ms**
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.210 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.226 ms**
5 ANN10 + T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.952 ms**
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.164 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.422 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.358 ms**
6 T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 2.370 ms**
7 ANN10 + T/P/ϕ DeepONet10	baseline	1.72	0.836	120	3.17e+03	16.6	271	0.621	3.01	0.949	1.03e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.414 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 68

T_0 = 1086.3 K P_0 = 3.702 atm ϕ_0 = 1.168 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 68 | NRMSE (util.py)

T0 = 1086.3 K P0 = 3.702 atm phi0 = 1.168 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.008 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.026 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 2.764 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.006 ms**
3 ANN10 + P-only DeepONet10	baseline	0.295	1.67	0.514	0.978	5.87	0.866	0.624	2.66	163	4.49e-16	Complete
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.010 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.422 ms**
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.634 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.306 ms**
5 ANN10 + T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.880 ms**
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.176 ms**
6 T/P/ϕ DeepONet10	baseline	1.01	1.81	2.29	1.08e+03	26.3	575	0.851	7.61	18.2	4.49e-16	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 1.404 ms**
6 T/P/ϕ DeepONet10	softlong	0.135	1.04e+08	4.17e+27	3.47e+25	5.67e+11	4.80e+21	1	2.05e+27	1.84e+27	1.22e-03	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.872 ms**
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.246 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 69

T_0 = 935.0 K P_0 = 2.045 atm ϕ_0 = 0.523 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

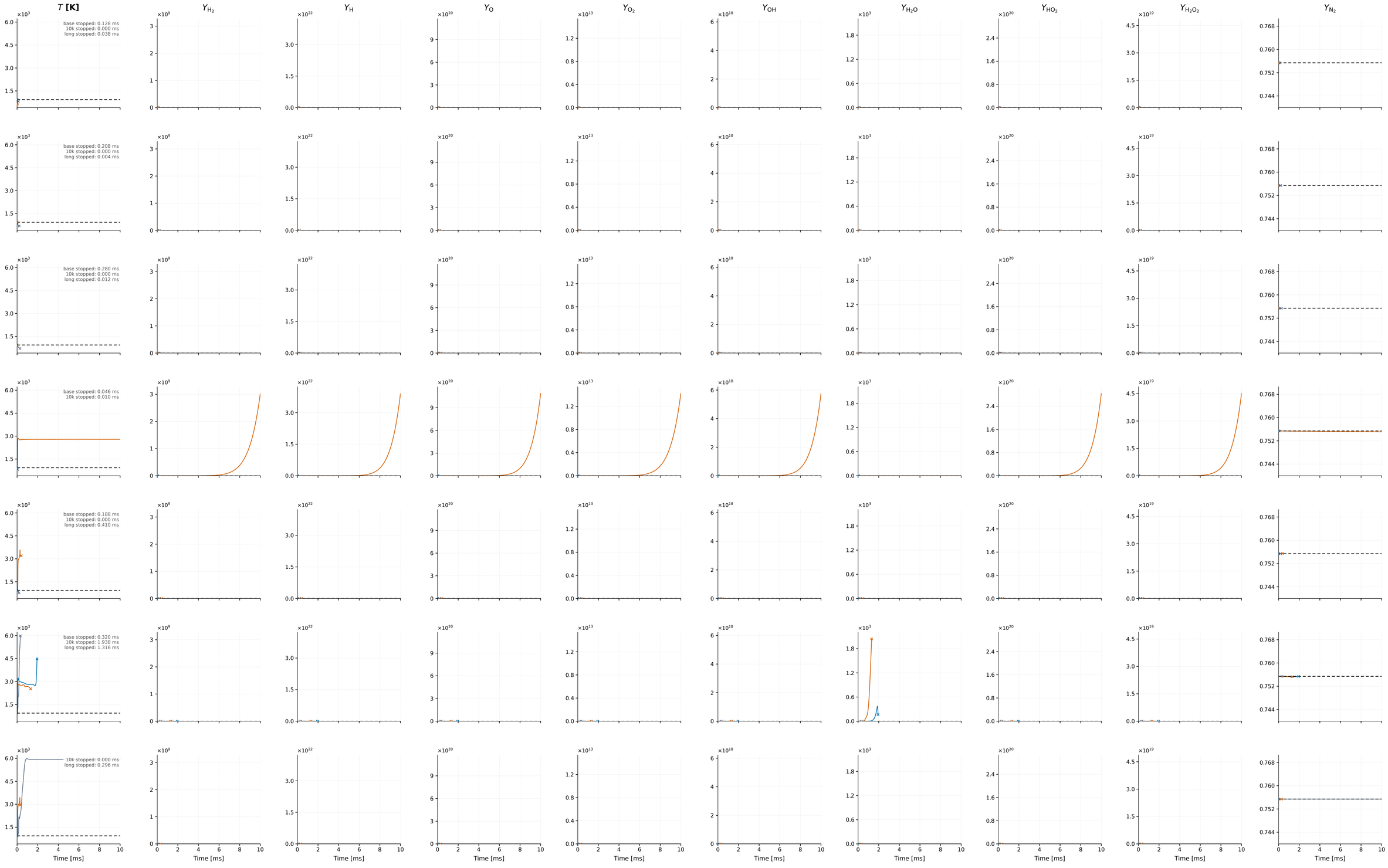
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 69 | NRMSE (util.py)

T0 = 935.0 K P0 = 2.045 atm phi0 = 0.523 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

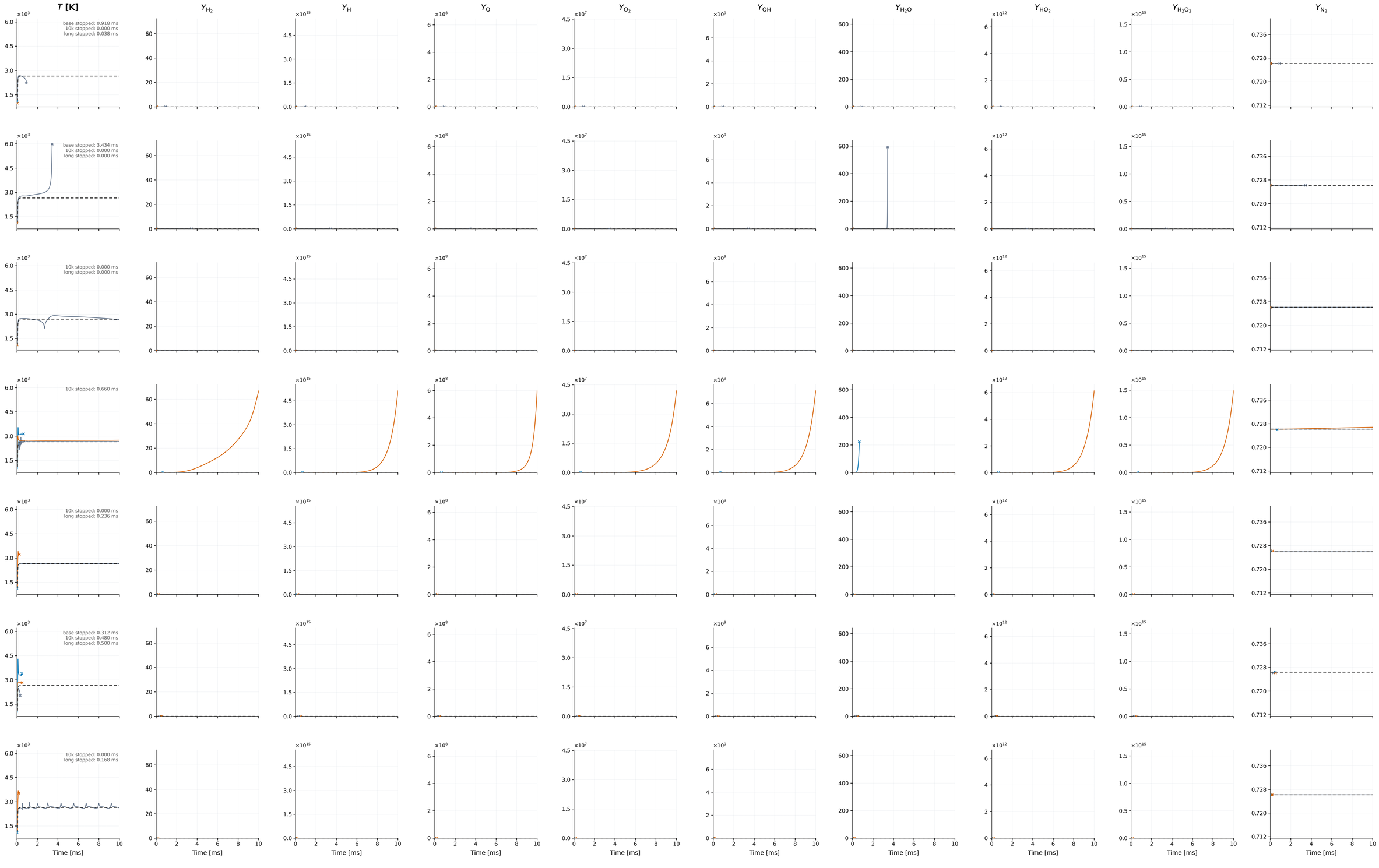
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.128 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.038 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.208 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.004 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.280 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.012 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.046 ms**
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.010 ms**
4 T/P DeepONet10	softlong	1.98	4.52e+10	2.67e+33	2.43e+31	1.38e+13	3.44e+29	1.88e+05	2.67e+26	1.10e+26	2.57e-04	Complete
5 ANN10 + T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.188 ms**
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.410 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.320 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 1.938 ms**
6 T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 1.316 ms**
7 ANN10 + T/P/ϕ DeepONet10	baseline	5.19	0.582	9.65e+08	1.69e+11	12.3	4.23e+11	2.17e+06	829	349	1.03e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.296 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 73

T_0 = 1106.0 K P_0 = 1.911 atm ϕ_0 = 1.915 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 73 | NRMSE (util.py)

T0 = 1106.0 K P0 = 1.911 atm phi0 = 1.915 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.918 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.038 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 3.434 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	baseline	0.06	0.5	0.694	2.01	0.0571	0.946	0.199	0.207	0.307	7.64e-16	Complete
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	baseline	0.0213	0.271	0.332	0.788	0.495	0.316	0.107	1.23	1.2	7.64e-16	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.660 ms**
4 T/P DeepONet10	softlong	0.05	864	1.10e+18	3.61e+11	4.92e+08	3.59e+11	0.979	1.24e+17	6.44e+20	4.76e-04	Complete
5 ANN10 + T/P DeepONet10	baseline	0.0125	0.0563	0.39	0.725	0.384	0.262	0.213	0.953	1.05	7.64e-16	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.236 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.312 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.480 ms**
6 T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.500 ms**
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.0314	0.167	0.638	1.09	0.399	0.587	0.172	0.993	1.07	7.64e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.168 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 74

T_0 = 1222.9 K P_0 = 4.971 atm ϕ_0 = 1.766 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 74 | NRMSE (util.py)

T0 = 1222.9 K P0 = 4.971 atm phi0 = 1.766 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.666 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.342 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.008 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.354 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	baseline	0.0146	0.0312	0.19	0.55	0.598	0.207	0.0463	1.03	0.965	1.83e-15	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.010 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.192 ms**
5 ANN10 + T/P DeepONet10	baseline	0.0414	0.092	0.578	0.866	0.539	0.599	0.208	1.07	0.987	1.37e-15	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.436 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.012 ms**
6 T/P/ϕ DeepONet10	softlong	0.116	1.59e+11	4.52e+29	2.97e+28	9.60e+14	1.64e+24	0.999	7.48e+29	6.75e+29	2.11e-03	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.678 ms**
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 76

T_0 = 1523.6 K P_0 = 3.626 atm ϕ_0 = 1.349 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Row 1: base stopped: 1.202 ms; 10k stopped: 0.000 ms; long stopped: 0.000 ms

Row 2: base stopped: 0.718 ms; 10k stopped: 0.000 ms; long stopped: 0.020 ms

Row 3: base stopped: 4.378 ms; 10k stopped: 0.000 ms; long stopped: 0.000 ms

Row 4: 10k stopped: 0.282 ms; long stopped: 0.182 ms

Row 5: 10k stopped: 0.000 ms; long stopped: 0.000 ms

Row 6: base stopped: 0.612 ms; 10k stopped: 0.292 ms; long stopped: 0.336 ms

Row 7: 10k stopped: 0.000 ms; long stopped: 0.000 ms

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 76 | NRMSE (util.py)

T0 = 1523.6 K P0 = 3.626 atm phi0 = 1.349 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

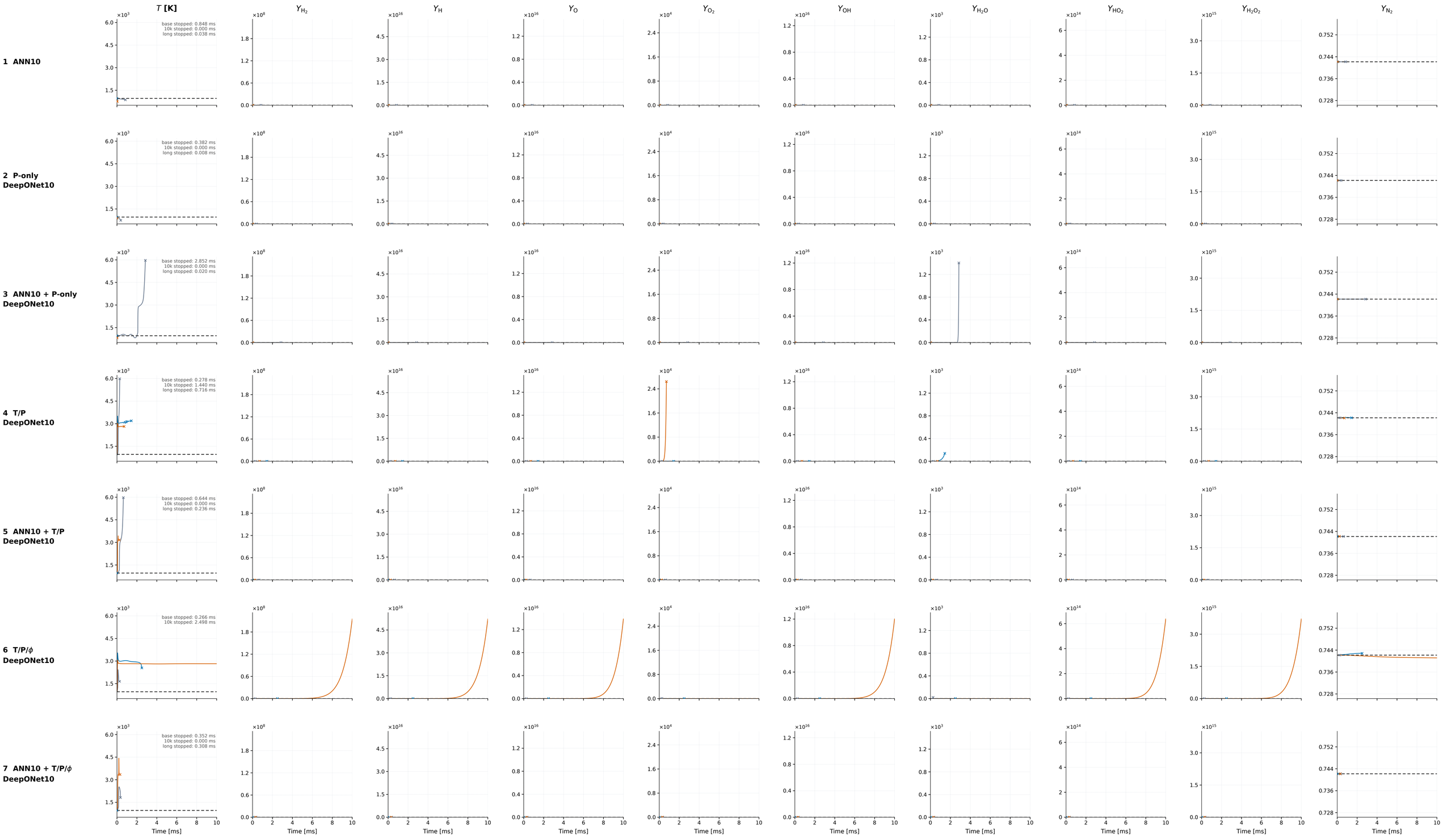
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.202 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.718 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.020 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 4.378 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	baseline	0.0553	0.0153	0.391	0.634	0.466	0.448	0.0778	0.653	0.534	6.02e-16	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.282 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.182 ms**
5 ANN10 + T/P DeepONet10	baseline	0.0546	8.51e-03	0.396	0.647	0.485	0.475	0.0889	0.671	0.58	6.02e-16	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.612 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.292 ms**
6 T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.336 ms**
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.714	2.11	1.21	590	89.7	351	0.664	27.1	163	6.02e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 78

T_0 = 962.3 K P_0 = 2.443 atm ϕ_0 = 1.140 | Raw self-fed rollout, $\Delta t = 2\,\mu$s





Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 78 | NRMSE (util.py)

T0 = 962.3 K P0 = 2.443 atm phi0 = 1.140 | Raw self-fed, 10 ms

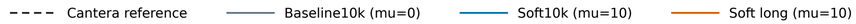
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

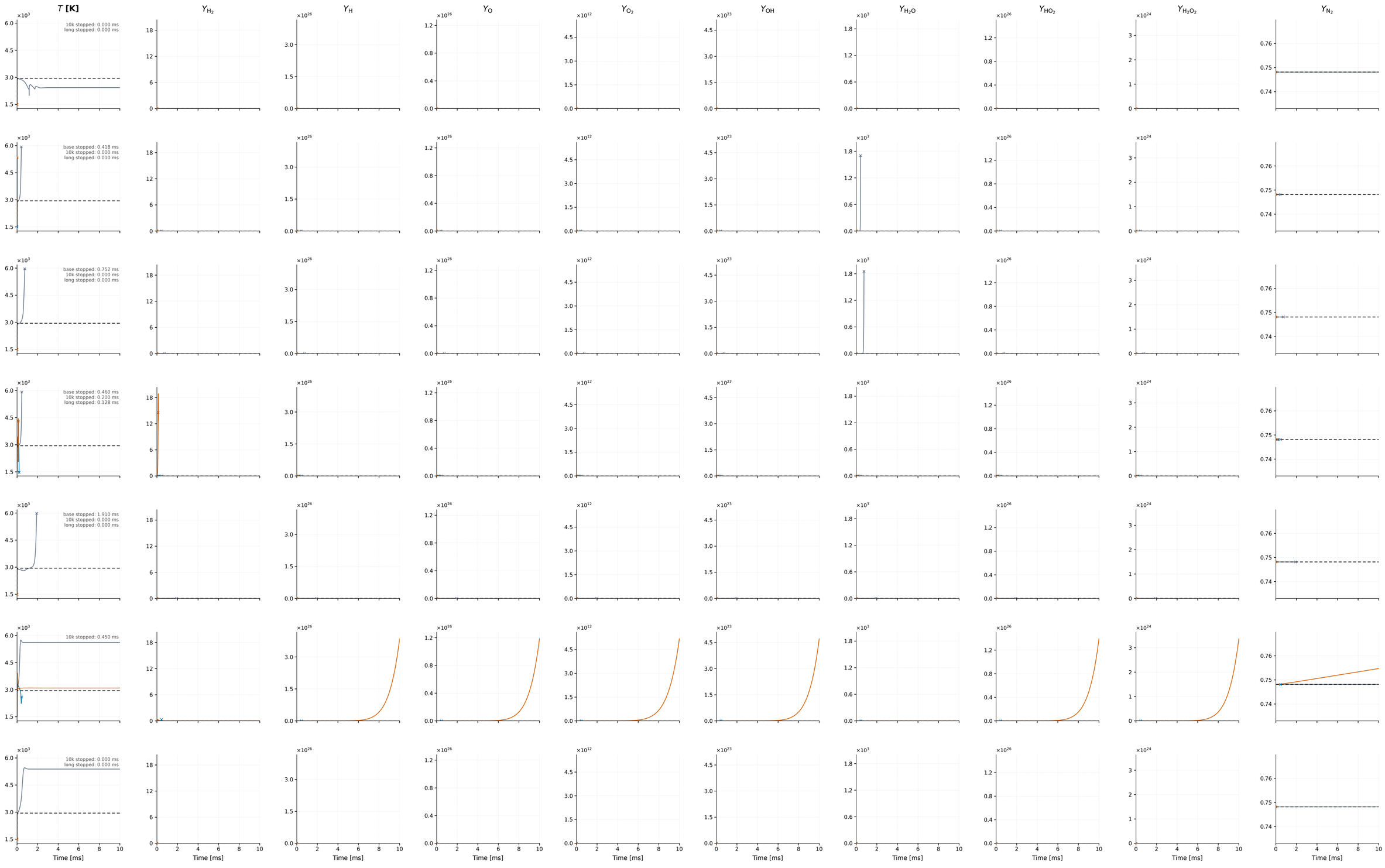
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.848 ms**
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.038 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.382 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.008 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 2.852 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.020 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.278 ms**
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 1.440 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.716 ms**
5 ANN10 + T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.644 ms**
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.236 ms**
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.266 ms**
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 2.498 ms**
6 T/P/ϕ DeepONet10	softlong	1.93	1.42e+09	5.74e+25	9.65e+24	0.937	1.73e+25	5.66e+03	2.77e+19	1.66e+20	9.12e-04	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.352 ms**
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.308 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 88

T_0 = 1502.1 K P_0 = 4.980 atm ϕ_0 = 0.859 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 88 | NRMSE (util.py)

T0 = 1502.1 K P0 = 4.980 atm phi0 = 0.859 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.165	2.68	0.929	0.95	0.951	0.9	0.71	0.956	0.948	8.90e-16	Complete
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.418 ms**
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.010 ms**
3 ANN10 + P-only DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.752 ms**
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.460 ms**
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.200 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.128 ms**
5 ANN10 + T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.910 ms**
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
5 ANN10 + T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	baseline	0.896	3.88	2.42	341	53.6	140	0.713	5.19	42.7	8.90e-16	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.450 ms**
6 T/P/ϕ DeepONet10	softlong	0.0502	16.4	2.15e+29	5.61e+27	3.16e+13	4.91e+24	1	1.10e+30	3.49e+29	4.97e-03	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.807	5.08	5.33	500	48.7	88	0.136	6.07	29.2	8.90e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.